

```
1  #include "Bank.h"
2
3  Bank::Bank() {
4      balance = 0.0;
5      depositNum = 0;
6      withdrawlNum = 0;
7      annualInterest = 0.02f;
8      monthlyCharges = 0.0;
9      serviceCharge = 0.0;
10
11 }
12
13 Bank::Bank(float b, int d, int w, float a, float m) {
14     balance = b;
15     depositNum = d;
16     withdrawlNum = w;
17     annualInterest = a;
18     monthlyCharges = m;
19     serviceCharge = 0.0;
20
21 }
22
23 void Bank::setBalance(float b) {
24     balance = b;
25
26 }
27
28 void Bank::setCharges(float m) {
29     monthlyCharges = m;
30 }
31
32 void Bank::setDeposit(int d) {
33     depositNum = d;
34 }
35
36 void Bank::setInterest(float a) {
37     annualInterest = a;
38 }
39
40 void Bank::setWithdrawl(int w) {
41     withdrawlNum = w;
42 }
43
44 float Bank::getBalance()const {
45     return balance;
46 }
47
48 int Bank::getDeposit()const {
49     return depositNum;
```

```
50 }
51
52 int Bank::getWithdrawl()const {
53     return withdrawlNum;
54 }
55
56 float Bank::getInterest()const {
57     return annualInterest;
58 }
59
60 float Bank::getCharges()const {
61     return monthlyCharges;
62 }
63
64
65 void Bank::deposit() {
66     float d{};
67
68     cout << "Enter an Amount to Deposit: ";
69     cin >> d;
70
71     while (d < 0)
72     {
73         cout << "ERROR: Must be greater than zero\n\n"
74             << "Enter amount to Deposit: ";
75         cin >> d;
76     }
77     balance += d;
78     depositNum++;
79 }
80
81
82 void Bank::withdraw() {
83     float w{};
84
85     cout << "Enter an Amount to Withdrawl: ";
86     cin >> w;
87
88     while (w < 0)
89     {
90         cout << "ERROR: Must be greater than zero\n\n"
91             << "Enter amount to withdrawl : ";
92         cin >> w;
93     }
94     balance -= w;
95     withdrawlNum++;
96 }
97
98 void Bank::calcInt() {
```

```
99     float interestM = annualInterest / 12;
100
101     interestM = balance * interestM;
102
103     balance = balance + interestM;
104 }
105
106 void Bank::monthlyProc() {
107     balance -= serviceCharge;
108
109     calcInt();
110
111     withdrawlNum = 0;
112     depositNum = 0;
113     monthlyCharges = 0;
114 }
115
116 void Bank::display()
117 {
118     cout << fixed << setprecision(2);
119     cout << "Withdrawals:\t" << withdrawlNum
120         << "\nDeposits:\t\t" << depositNum;
121     monthlyProc();
122     cout << "\nService Charges:\t" << serviceCharge
123         << "\nEnding Balance:\t" << balance;
124
125 }
```